

1 Is Purity of Pure consistent with Bacon’s Logical Combinatorialism?

1.1 State-of-project notes

Jeremy Goodman, July 2026. Developed in extended collaboration with Claude (Anthropic). These notes are self-contained: all non-standard terminology is defined here. Each result carries a verification status; “checked $n\times$ ” means the derivation has been hand-verified n times within the collaboration but not independently refereed. Intended audience: colleagues and AI research agents joining the project.

Primary sources. - Andrew Bacon, “Logical Combinatorialism”, *Philosophical Review* 2020. <https://andrew-bacon.github.io/papers/Logical%20Combinatorialism.pdf> - Andrew Bacon & Cian Dorr, “Classicism” (2024). <https://andrew-bacon.github.io/papers/Classicism.pdf>

1.2 1. The problem

Bacon’s paper develops a theory, in classical higher-order logic, of three interlocking notions: **logical necessity** ($\Box$), **fundamentality** (predicates Fun_σ , for each type σ), and **purity** (predicates Pure_σ) — where the pure entities are, intuitively, the purely logical ones (negation, conjunction, identity, ...), and the fundamental entities are the metaphysically basic non-logical ones. His central principle:

- **Quantified Logical Necessity (QLN).** For pure X and pairwise-distinct fundamental $\bar{z}$: $\Box X(\bar{z}) \leftrightarrow \forall \bar{x} X(\bar{x})$. We distinguish its two directions:
 - **Recombination** ($\Box X(\bar{z}) \rightarrow \forall \bar{x} X(\bar{x})$; equivalently, any instantiated pure condition possibly holds of the fundamental entities). Philosophically essential to the package.
 - **Exhaustion** ($\forall \bar{x} X(\bar{x}) \rightarrow \Box X(\bar{z})$; equivalently, any possible pure pattern of the fundamental entities is actually instantiated). More contentious; it entails that $\top$ and $\perp$ are the only pure propositions (§4, T1). The Bacon-Dorr “Classicism” paper has models of the Recombination direction with many pure propositions, so we track which results need Exhaustion.

From QLN (Recombination direction only) Bacon derives:

- **Quantified Separated Structure (QSS).** For pure X , Y and pairwise-distinct fundamental $\bar{z}$: $X(\bar{z}) = Y(\bar{z}) \rightarrow X = Y$. (“Evaluation at a fundamental tuple is injective on pure operators.”)

In a footnote (fn. 59 of the published version), Bacon proves that the following two principles are *jointly inconsistent* with the above plus the existence of a fundamental proposition:

- **Purity of Pure (PP).** $\text{Pure}(\text{Pure}_\sigma)$ — the property of being pure is itself pure.
- **Purity of Fun.** $\text{Pure}(\text{Fun}_\sigma)$.

The proof is a liar-style diagonal. Bacon leaves open, and we investigate:

The Question. Is PP consistent with the rest of the theory plus the existence of a fundamental proposition (Purity of Fun *not* assumed)?

Unless stated otherwise, results are for the case of **exactly one fundamental entity, a proposition r** (nothing fundamental at any other type). “PP at σ ” means the single instance asserting the purity of the purity-predicate for type- σ entities; so “PP at $t \rightarrow t$ ” asserts that $\text{Pure}_{\{t \rightarrow t\}}$ — the predicate applying to pure *operators* — is itself pure.

1.3 2. The background theory

To (core theory). Bacon's system H (propositional tautologies, modus ponens, generalization, universal instantiation, $\beta\eta$ -equivalence as a biconditional axiom at type t), plus: - **Rule of Equivalence:** from a theorem $A \leftrightarrow B$ infer $A = B$ (applied to open formulas as well; this is how Bacon secures Booleanism — the identification of Boolean-equivalent propositions). - **Modalized Functionality:** $\Box \forall x (Xx = Yx) \rightarrow X = Y$. (Note: *plain* Functionality, $\forall x (Xx = Yx) \rightarrow X = Y$, is **not** assumed, though it is true in Bacon's models. Likewise the Barcan formula is true in his models but not in the theory.) - $\Box$ is defined as $(= \top)$. - **Purity schema:** every closed term containing only logical vocabulary denotes something pure. - **Application closure:** $\text{Pure}(X) \wedge \text{Pure}(y) \rightarrow \text{Pure}(Xy)$. (With the Purity schema this yields: any closed term whose constants are all pure denotes something pure — via Bacon's device of abstracting the constants into a closed λ -term.) - **Persistence** (used only where flagged): $\text{Pure}(x) \rightarrow \Box \text{Pure}(x)$.

Bacon's appendix model (used for all model-theoretic results). Substitutions form the monoid M = finite sequences of natural numbers under concatenation; propositions are arbitrary sets of substitutions, $D_t = P(M)$, with $i \cdot p = \{j : j \cdot i \in p\}$; a proposition is true iff it contains the empty sequence (the "root"). Function domains contain exactly the maps respecting the substitution action in the sense of his Def. 7.2. Pure_σ is interpreted as applying (at each substitution) to the denotations of closed terms with no non-logical constants; Fun_t as applying to the denotations of the constants (here: just r). An element is **invariant** if every substitution fixes it. Useful picture: a substitution i is a possibility at which each entity x plays the role actually played by $i \cdot x$.

1.4 3. Central definitions

All at type t / $t \rightarrow t$ unless noted; everything generalizes by type.

- **fun'(x)** := for all pure Y, Z of type $t \rightarrow t$: $Yx = Zx \rightarrow Y = Z$. ("Evaluation at x is injective on pure operators." This is the *pure surrogate of fundamentality*: QSS says fundamental propositions are **fun'**, but **fun'** is definable — and, **given PP, fun' is itself pure**, since its only non-logical constant is **Pure**. This self-applicative fact drives everything below.)
- **Reversible** (invertible): Z is reversible iff for some pure Z' , $Z \circ Z' = \text{id} = Z' \circ Z$. **G** := the set of pure reversible operators — a group under composition. Known members: the identity id and negation $\neg$; more generally, for any pure proposition A , the operator $\lambda p.(p \leftrightarrow A)$ (call these **biconditional operators**; they compose by $p \leftrightarrow A \leftrightarrow B$ and are each their own inverse). If $\top, \perp$ are the only pure propositions, the biconditional operators are just id and $\neg$.
- **X $\approx$ Y** := $X = Y \circ Z$ for some Z in **G**. (Equivalence relation: "same up to invertible pure re-description of the input".)
- **Kinds**. The kind of a pure operator X is its $\approx$ -class. A proposition p is of kind K iff $p = X(q)$ for some X in K and **fun'** q . Under L2 (below), no proposition is of two kinds, and every kind is realized ($X(r)$ is of X 's kind).
- Two candidate transfer/classification principles, currently **conjectures**, that the refutation results consume:
 - **L2 (transfer)**. For pure X, Y and **fun'** p, q : $Xp = Yq \rightarrow X \approx Y$. Strong form (used where flagged): the $\approx$ -witness also relates the inputs — $\exists Z \in G (X = Y \circ Z \wedge q = Zp)$.
 - **Classification of G**. Any of the following, in decreasing strength:
 - * **Inv**: $G = \{\text{id}, \neg\}$.
 - * **WI**: every member of G is a biconditional operator $\lambda p.(p \leftrightarrow A)$.
 - * **TU (truth-uniformity)**: every member of G either preserves the truth-values of all propositions or flips the truth-values of all propositions. (Follows from "pure invertible operators are truth-functional".)
 - * **RS (rigid specification)**: there exists a pure, instantiated property G^* applying only to **fun'** propositions such that no member of G maps one G -instance to a distinct* G -instance. ($\text{Inv} \vdash \text{RS}$ and $\text{TU} \vdash \text{RS}$, witness $G :=$ "true and fun"; RS is

choice-flavored where the others are classification-flavored.)

- **PC (pure completeness).** For any plurality of pure entities of a given type, there is a pure property applying to exactly them (the infinite disjunction of their identity-properties, assumed pure). Motivated by Bacon’s fn. 60; accepted by JG as a desideratum. In finite cases PC is a theorem (finite disjunctions of pure things are pure by application closure); it has content only when some type has infinitely many pure entities.

1.5 4. Object-language theorems

Assumed throughout §4: $T_0 + PP$ at $t \rightarrow t$ + the existence of a fun' proposition (write r for a witness; when “fundamental” is said, QSS supplies $\text{fun}'(r)$). Extra assumptions flagged per item.

T1 (Triviality of pure propositions). [Needs the Exhaustion direction of QLN.] Every pure proposition is $\top$ or $\perp$. (*Checked repeatedly.*) Consequently, in the full theory the biconditional operators are exactly $\{\text{id}, \neg\}$, and WI collapses into Inv there.

T2 (Basic fun' facts). (*Checked repeatedly; no QLN needed beyond QSS-at-r.*) $\neg \text{fun}'$ is closed under negation, and more generally under every member of G . $\neg \text{fun}'(\top)$, $\neg \text{fun}'(\perp)$; hence $r \neq \top$, $\perp$, and $r \neq \neg r$. $\neg$ For every pure p : $\Diamond(r = p)$. Hence $\Diamond \text{Pure}(r)$ [uses Persistence], and “ r is non-contingent” is a proposition that is provably false but provably possible. $\neg$ The six propositions $\top$, $\perp$, r , $\neg r$, $(r = \top)$, $(r = \perp)$ are pairwise distinct.

T3 (Hereditry). [Needs necessitated QSS and Persistence.] $\Diamond \text{Fun}(x) \rightarrow \text{fun}'(x)$: whatever possibly plays the fundamental role is fun' . (*Checked 2 $\times$.*) The converse fails in Bacon’s model (§5), but is a candidate axiom (“ $\text{fun}' \leftrightarrow \Diamond \text{Fun}$ ”) whose consistency is open.

T4 (No definable higher-type fun' entities). For every closed pure C of type $t \rightarrow (t \rightarrow t)$: $\neg \text{fun}'_{\{t \rightarrow t\}}(C(r))$. Nothing metaphysically definable from r is a fun' entity one type up. (*Checked 2 $\times$.*) Proof: the diagonal $D_C := \lambda q. \neg(Cqq)$ is closed and pure; $C(r) = D_C$ would give $Crr = \neg Crr$; so the pure properties “ $= D_C$ ” and “ $\lambda X. \perp$ ” witness the failure of injectivity at $C(r)$. Consequence: fun' entities at type $t \rightarrow t$, if any, cannot be generated from the propositional fundamentalia — they would have to be independently fundamental.

T5 (Proliferation). $T_0 + PP + \text{fun}'(r)$ refutes “the fun' propositions are exactly r and $\neg r$ ”: there is a fun' proposition distinct from both. (*Checked 2 $\times$.*) So PP forces the class of fun' propositions beyond the fundamental proposition and its negation. (Proved by running the liar of T6 with the two-element hypothesis in place of L2 + classification; the relation between r and $\neg r$ is a *known* member of G , which is all that case needs.)

T6 (The liar; main refutation results). Write $D := \lambda p. \forall X \forall q. (\text{Pure}(X) \wedge \text{fun}'(q) \wedge p = Xq \rightarrow \neg Xp)$ — pure, by PP. Then, in increasing generality: $\neg T_0 + PP + \exists \text{fun}' + L2 + \text{Inv} \vdash \perp$. (*Checked 2 $\times$.*) $\neg T_0 + PP + \exists \text{fun}' + L2 + \text{WI} \vdash \perp$, via a “master equation” indexed by the pure propositions: writing $a_A := D(d \leftrightarrow A)$ for pure A and $d := Dr$, one derives — for all pure A : $a_A \leftrightarrow \forall C. (\text{Pure}(C) \rightarrow (a_C \leftrightarrow \neg A))$ — and this family is propositionally inconsistent. (*Checked 3 $\times$.*) No assumption on the number of pure propositions; compatible with the Recombination-only setting. $\neg T_0 + PP + \exists \text{fun}' + L2 + \text{TU} \vdash \perp$. (*Checked 1 $\times$.*) TU is currently the weakest classification principle known to suffice on this route. $\neg T_0 + PP + \text{strong-L2} + \text{RS} \vdash \perp$. (*Checked 1 $\times$.*) A different route: the existential-form diagonal $B := \lambda x. \exists Y \exists y. (\text{Pure}(Y) \wedge G(y) \wedge x = Yy \wedge \neg Yx)$ with G the rigid specification; two-step contradiction. RS is not comparable to TU; both are entailed by Inv.

Contrapositives (the current standing of the Question). Bacon’s theory + one fundamental proposition + PP proves $\neg(L2 \wedge \text{TU})$ and $\neg(\text{strong-L2} \wedge \text{RS})$. So: if L2 and any one of the classification principles can be independently secured (derived from the theory, or philosophically motivated as part of the intended conception of purity), PP is refuted and the Question is settled negatively. Conversely, any model of the theory + PP must contain either cross-input collisions of pure operators on fun' propositions ($\neg L2$) or pure invertible

operators of a non-classical character ($\neg$ TU: neither uniformly truth-preserving nor uniformly truth-flipping).

T7 (Absorption; why the liar alone does not suffice). From $T_0 + PP + \exists \text{fun}' + L2$ *alone* (no classification principle): the liar family $\{a_g : g \in G\}$ satisfies $\neg a_{\text{id}}$ together with $\exists g \in G a_g$. (*Checked 1 $\times$.*) Informally: without a classification of G , this argument yields no contradiction — for all we know, the diagonal is *absorbed* by some pure invertible about whose action nothing further is provable. A companion result at the level of kinds (needing PP also at type $(t \rightarrow t) \rightarrow t$) shows the diagonal-on-kinds likewise has no fixed point but a shifted one. (*Checked 1 $\times$.*) Moral: **on this strategy, the remaining distance to a refutation of PP is a classification of G** (other strategies for proving inconsistency may of course exist); group-theoretic generalities cannot bridge the gap (every group acts transitively on itself, and that is not enough).

T8 (Growth). Five pairwise-distinct kinds are provable outright: those of id , $\Box$ (necessity), $\Diamond$ (possibility), $\lambda p. \top$, $\lambda p. \perp$ — distinctness needs no classification principle. Under L2, kind-uniqueness holds, and the map sending each nonempty set S of kinds to the pure property “being of some kind in S ”, then evaluating at r , is injective. Hence $T_0 + PP + \exists \text{fun}' + L2$ proves $\geq 2^5 - 1 = 31$ pairwise distinct propositions and 31 pairwise distinct pure operators of type $t \rightarrow t$. (*Checked 2 $\times$.*) **Caveat:** an earlier write-up proving full infinitude of D_t used the axiom refuted in T5 and is therefore superseded (its lemmas remain useful); unbounded growth from the 31-bound requires showing the new kinds outrun the collapsing action of G , which is again the classification problem. Under PC the story sharpens (next item).

T9 (Cardinality dichotomy under PC). Heuristic but precise at the level of counting. With $\kappa :=$ the number of kinds: PC makes “being of some kind in S ” pure for *arbitrary* sets S of kinds, so 2^κ many pairwise-distinct pure operators exist; each $\sim$ -class has size at most $|G|$ (constant operators, for instance, are fixed by the whole action); hence $2^\kappa \leq \kappa \cdot |G|$, i.e. **either κ is finite or $|G| \geq 2^\kappa$** . So a model of the theory $+ PP + L2 + PC$ must have either finitely many kinds (under pressure from T8, though not yet excluded) or a group of pure invertibles that exponentially dominates the kinds it acts on. This packages the whole problem: exclude finite κ , cap $|G|$, and PP is refuted; conversely a consistency proof must exhibit one of the two exotic shapes.

1.6 5. Model-theoretic results (Bacon’s appendix model)

M1 (Where PP fails, and why). *PP holds* at the bottom type: the invariant propositions are exactly $\{\top, \perp\}$, and the interpretation of Pure t computes out to the non-contingency operator $\lambda p. (\Box p \vee \Box \neg p)$ — a closed logical term. $\bar{PP}$ first fails at type $t \rightarrow t$, necessarily so: the model verifies Purity of Fun, QSS, and a fundamental proposition, so by Bacon’s fn.-59 argument it cannot also verify PP there. The obstruction is Tarski-style: “being the denotation of a constant-free closed term” is not itself so denotable. Corollary: the fn.-60 argument for PP fails *in Bacon’s own model* — the infinite disjunction of pure identity-properties exists (domains are complete Boolean algebras) but is not pure; the step “infinitary disjunctions of pure things are pure” is exactly what breaks. (*Checked 2 $\times$.*)

M2 (The invariance reading of purity is not an option). For every set T of propositions, the operator $G_T := \lambda p. \{i : i \cdot p \in T\}$ is a well-defined, invariant element of the type- $t \rightarrow t$ domain, and $T \mapsto G_T$ is injective; so there are more invariant operators than propositions, and evaluation-at- r cannot be injective on them. Explicit failure of QSS under the invariance reading: $G_{\{\{x_0\}\}}$ (x_0 any proposition outside the orbit of r — the countable set $\{i \cdot r : i \in M\}$ of things r can be sent to) and $\lambda p. \perp$ are distinct, invariant, and agree at r . (*Checked repeatedly.*) Conversely every invariant operator is of the form G_T . So the identification of purity with invariance contradicts QSS (given a fundamental proposition), and exotic invariant operators are unavoidable background structure; the live question is only ever which invariants are *certified* pure.

M3 (fun' = free/generic). In the model: $\text{fun}'(p)$ iff $\forall p \neq \perp$ for every definable operator $V \neq \lambda q. \perp$ — iff p satisfies no non-trivial pure law (“ p is a free generator relative to the definable operators”). Consequences: every fun' proposition must be possibly-necessary and possibly-impossible; the fun' propositions form a topologically meager class (generic in the algebraic sense, not the Baire sense); Bacon’s model-construction (his Theorem 10.1 gluing) is precisely a method for building such generic elements, which is *why* his r is fun' . (*Checked $2\times$, modulo the fact that Theorem 10.1 itself is stated without proof in the paper.*)

M4 (fun' outruns the fundamental). Preimages of fun' propositions under any substitution are fun' (“heredity”, the model face of T3). Explicit fun' proposition not related to r by any pure invertible: write the pattern of r into the region above one substitution and leave everything else empty; the result is fun' (heredity) but cannot purely recover r , since from other vantage points it has become $\perp$ while r has not, and pure operators cannot manufacture information their argument lacks. (*Checked $2\times$.)* Consequently “ fun' = the pure-invertible images of the fundamental” is false in the model. Also refutable, in two independent ways: interpreting Fun “widely” so that its extension contains one representative of each $\approx$ -class of fun' propositions — (i) if p, q are distinct fundamental propositions then $p \wedge q$ is provably fun' but purely definable from the pair, and a contradiction with Separated Structure follows for the triple; (ii) in the single-fundamental case the proposal undershoots: the $\approx$ -classes of r and $\neg r$ do not exhaust the fun' propositions (this item). What survives: the modal thesis $\text{fun}' \leftrightarrow \Diamond \text{Fun}$ (the direction $\Diamond \text{Fun} \rightarrow \text{fun}'$ is theorem T3; the direction $\text{fun}' \rightarrow \Diamond \text{Fun}$ is open — it looks unrealizable in natural constructions of the model but is not refuted).

M5 (The classification principles are not entailed by the PP-free theory). The key construction, in full. Call $i \cdot p$ the **view of p from i** — p ’s pattern above the substitution i , read as a proposition in its own right; a proposition is true at i just in case its view from i contains the actual world. Fix one non-actual world w (a sequence of length ≥ 1 , say $\langle 5 \rangle$), and let $s :=$ the proposition true at exactly w , and $s' :=$ the proposition true at exactly w and the actual world. So s and s' differ only at the actual world; s is false, s' is true. Recall from M2 that every set T of propositions determines an invariant operator $G_T := \lambda p. \{i : i \cdot p \in T\}$, and that taking $T_0 :=$ the set of *true* propositions gives $G_{T_0} =$ the identity. Define **$f := G_T$ where T is T_0 with the membership of s and s' exchanged** (s added, s' removed). Unwinding the definitions: $f(p)$ agrees with p in truth-value at every substitution except those from which p ’s view is s or s' , where it disagrees. Now the verification: - *No echo*: from any substitution other than the actual world, the view of s (equally of s') is either the empty proposition or the proposition true at exactly the actual world — because the only prefixes of w are the empty sequence and w itself — and neither of these is s or s' . So the only substitution from which s or s' has a view inside the pair $\{s, s'\}$ is the actual world. - *f transposes the pair*: by “no echo”, the set of substitutions at which f flips s is exactly $\{\text{the actual world}\}$, so $f(s) = s'$; symmetrically $f(s') = s$. - *f is an involution*: a short case analysis (the flagged audit point of this construction) shows that no proposition outside the pair is mapped by f into the pair; together with the previous item, the set of substitutions at which f flips p is therefore the same for p and for $f(p)$, which gives $f(f(p)) = p$ for every p . So f is invertible — its own inverse. - *f fixes $\top$ and $\perp$* (no view of $\top$ or $\perp$ is ever s or s'). So f preserves the truth-value of $\top$ while taking the true s' to the false s : f is neither uniformly truth-preserving nor uniformly truth-flipping — **TU fails for f** . - *f is not a biconditional operator*: a biconditional operator $\lambda p. (p \leftrightarrow A)$ other than the identity moves *every* proposition ($p \leftrightarrow A = p$ forces $A = \top$), whereas f fixes $\top$. So **WI fails for f** , hence so does Inv. - *Certifying f pure*. f is invariant (M2), so it is at least a candidate. Simply adding f to the extension of Pure over the existing model yields a contradiction with QSS: choose the pair $\{s, s'\}$ outside the orbit of r (possible, since the orbit is countable and there are uncountably many valid choices of w -based pairs); then no view of r is ever in the pair, so $f(r) = r = \text{id}(r)$ with $f \neq \text{id}$, contradicting the injectivity of evaluation at r . The repair: treat f as a new item of *logical vocabulary* and re-run Bacon’s model construction (his Theorem 10.1) over the enlarged language, rebuilding the fundamental proposition so that it is generic with respect to the enlarged stock of definable operators, including those

built from f . Modulo Theorem 10.1 — which Bacon states without proof — this yields a model of the PP-free theory in which f is pure.

Conclusion: **WI and TU (and Inv) are genuinely additional commitments — the theory without PP does not prove them.** (*Construction checked 2×; the re-gluing step is plausible but unverified.*) Complementary positive tool, the **collision method**: the one *definable* candidate exotic invertible, $\lambda p.(p \leftrightarrow \text{“}p \text{ is non-contingent”})$, is provably non-injective already in $T_0 + \exists \text{fun}'$ — because “ r is non-contingent” is false-but-possible, and evaluating there collides with evaluating at $\top$. (*Checked 2×.*) Open: whether the collision method can be made to work against invertibles given only existentially (as PP-derived ones would be).

M6 (Supervenience yes, independence no). Distinct substitutions act differently on some fun' proposition (so everything about a possibility supervenes on the roles of the fun' propositions there). But joint realization of finite assignments fails twice over: orbits are countable while there are continuum-many propositions (so even a single proposition cannot be sent to an arbitrary target — note this means the *semantic* Fundamental Independence condition of Bacon’s §5 fails in his own model, harmlessly: what his construction actually delivers is “every consistent definable demand is realized”); and there are fun' propositions in distinct $\approx$ -classes standing in substitution-preserved relations (one strictly contains another), which no substitution can break. Moral: $\approx$ -inequivalence does not imply modal independence; the fun' propositions are potential role-occupants, not freely recombinable atoms. (*Checked 2×.*)

M7 (Fundamental Completeness). Bacon notes his model falsifies Fundamental Completeness (every entity is a pure operator applied to fundamental entities). Explicit witness: enumerate the closed terms and build the proposition whose view from the n -th branch of the model is the complement of the n -th term’s denotation there — a Tarski diagonal, definably-unreachable from r . Whether everything is reachable from r by *invariant* (not necessarily definable) operators is equivalent to injectivity of the map $i \mapsto i \cdot r$, which is construction-sensitive and open. (*Checked 2×.*)

1.7 6. Open problems and suggested attacks

Ordered roughly by expected value.

1. **Calibrate L2 in Bacon’s model.** Do two non- $\approx$ pure operators ever agree on fun' inputs there? The necessity/possibility case is provably collision-free (by comparing truth at the actual world: in the model, “possibly p ” is true whenever $p \neq \perp$, while “necessarily p ” is true only for $p = \top$, so the two can only collide degenerately). A systematic check over the definable operators would either refute L2 (removing the main refutation route’s premise, and teaching us what PP-models can exploit) or support seeking an object-language derivation of L2 from the theory (with or without PP). Concrete, finite-flavored casework; well-suited to mechanization.
2. **Cap or characterize G.** Everything funnels here (T7, T9). Sub-questions: can the collision method (M5) be schematized against existentially-given invertibles? Is “pure invertibles are truth-functional” ($\vdash \text{TU}$) derivable from any independently attractive granularity principle? Does the Bacon-Dorr Classicism setting (many pure propositions) prove, refute, or leave open TU/WI — and does it verify L2? Note for the Recombination-only setting: there may be many pure propositions, hence many biconditional operators, so even WI leaves G large; the master-equation version of T6 is the tool built for that regime, and RS is the classification-free alternative.
3. **Exclude finitely many kinds under PC** (T9’s first horn). If the five base kinds provably generate infinitely many kinds modulo the action of G — e.g., because the growth construction of T8 produces kinds faster than any group action can identify them — then T9 forces $|G| \geq 2^\kappa$ with κ infinite, and *any* independently motivated ceiling on G finishes the refutation.
4. **Prove consistency.** By T6’s contrapositive, any model of the theory + PP must falsify L2 or TU — it must contain either pure operators from distinct $\approx$ -classes agree-

ing on fun' propositions, or pure invertible operators that are neither uniformly truth-preserving nor uniformly truth-flipping. No such model is currently known; constructing one would settle the Question affirmatively. By M1-M2, it would have to differ from Bacon's substitution-structure models in a fundamental way (in particular, the interpretation of Pure would have to have its own denotation in its extension — a genuinely self-referential condition that Bacon's models do not satisfy above the bottom type).

1.8 7. Known error-prone zones (for auditors)

- Sign/negation bookkeeping around “inner” negation ($X \circ \neg$) versus “outer” negation ($\neg \circ X$): the single largest source of corrected errors in this project. Every liar-style derivation should be re-audited at exactly these steps.
- Truth at a substitution versus identity of propositions: in the model, r and $\top$ are distinct elements, yet the proposition ($r = \top$) is true at some substitutions — “true at i ” concerns the images $i \cdot r$ and $i \cdot \top$, not the elements r and $\top$ themselves. Conflating these two levels caused several retracted arguments.
- Bacon's models validate more than his theory (plain Functionality, the Barcan formula): arguments valid in the model need re-derivation before being claimed as theorems.
- Group-versus-monoid intuitions: substitutions do not cancel; stabilizer-style reasoning from group actions fails (e.g., “trivial stabilizer” does not imply injectivity of $i \mapsto i \cdot r$).
- Cardinality claims about “reachability”: the orbit of any proposition is countable; “fun” is an injectivity property and implies nothing about surjectivity of the reachable targets.